Exercise Sheet 01

Mistress Pell's Ravens

House rules.

- One file per exercise, in `my_work/`: `ex1_pay_maths.py`, `ex2_names.py`, …
- Start each file by typing the **Given** lines. Then write the rest yourself.
- Run it (`python my_work/ex1_pay_maths.py`) and compare with the **expected output**, character by character.
- Stuck for more than 10 minutes? Look at the cheat sheet, then skip and come back later.
- Hand in: commit and push to `lesson-01`, then open a pull request. Tell Lars which raven was the most annoying.

Exercise 1 — Pay maths

10 min · `ex1_pay_maths.py`

> **Raven from Mistress Pell (Paymaster):** *"The Grimoire failed again at the full moon. Run-time error '13', naturally. Can you redo the basic pay maths in Python? Yours in haste, P."*

GIVEN

```
annual_salary = 32500
weekly_hours = 37.5
part_time_hours = 22.5
holiday_days = 17
raise_percent = 3
years = 3
```

Calculate and print, each rounded to 2 decimal places where it makes sense:

(a) the monthly salary;

(b) the hourly rate (52 weeks a year);

(c) the salary of someone doing the same job for `part_time_hours` a week;

(d) how many full weeks and leftover days `holiday_days` are (use `//` and `%`);

(e) the salary after `years` yearly raises of `raise_percent` %, each on top of the last.

EXPECTED OUTPUT

```
Monthly salary: 2708.33
Hourly rate: 16.67
Part-time salary: 19500.0
Holiday: 2 weeks and 3 days
Salary after raises: 35513.63
```

Hint: "To the power of" is `**` in Python. If you use `^`, you'll meet it again in the Grimoire.

Exercise 2 Names and text

10 min · `ex2_names.py`

> **Raven from Orrin (Clockworks):** *"The Scriptorium's copy of the Roll is a mess. Dalvric's incantation 'cleaned' it by burning every second row. Please be gentler than Dalvric."*

GIVEN

```
raw_first = "  mAREN "
raw_last = "ashDOWN  "
raw_id = "  tm-0042 "
```

(a) Clean the names (no spaces around them, proper capitals) and print the full name.

(b) Print the initials with dots.

(c) Print the raven address: *first.last*@thornmere.realm, all lowercase.

(d) Clean the ID: no spaces, uppercase, no dash.

(e) Lady Siv's clerk is called `"Anne Marie"` `"Blackwood"`. Print her raven address, without the space.

EXPECTED OUTPUT

```
Maren Ashdown
M.A.
maren.ashdown@thornmere.realm
TM0042
annemarie.blackwood@thornmere.realm
```

Exercise 3 A pay scroll the Queen can love

10 min · `ex3_payslip.py`

> **Decree from Queen Aldara, sent at dusk:** *"The pay scrolls look 'a bit 1019'. Commas! Things lined up! Enjoy the rest day. A."*

GIVEN

```
name = "Maren Ashdown"
month = "March"
annual_salary = 32500
pension_rate = 0.05
tax_rate = 0.20
```

Basic pay is a twelfth of the salary. Guild dues are 5 % of basic pay. Crown tax is 20 % of (basic pay minus guild dues). Net pay is what's left. Print exactly this:

EXPECTED OUTPUT

```
PAY SCROLL: Maren Ashdown, March
---------------------------
Basic pay       2,708.33 cr
Guild dues        135.42 cr
Crown tax         514.58 cr
---------------------------
Net pay         2,058.33 cr

Guild dues rate: 5.0%
Annual salary: 32,500 cr
```

Hint: Each line is 27 characters: the label left-aligned in 15, the amount right-aligned in 12. Make the money text first: `basic_text = f"{basic:,.2f} cr"`, then `f"{'Basic pay':<15}{basic_text:>12}"`. And `"-" * 27` draws the line.

Exercise 4 Text that should be numbers

10 min · `ex4_conversions.py`

> **Raven from Mistress Pell:** *"The Scriptorium copies EVERYTHING out as text. Apparently '32,500.00 cr' is not a number. The Grimoire says 'Type mismatch'. I say 'please'."*

GIVEN

```
salary_text = "32,500.00 cr"
hours_text = " 37.5 "
bonus_text = "12.5%"
member_text = " Yes "
subject_number = 42
```

(a) Turn the first three into numbers and `member_text` into `True` / `False`. Print each value together with its `type()`.

(b) Build the subject ID: `TM` followed by the number padded to 4 digits.

(c) Print the bonus in crowns (salary × bonus rate).

EXPECTED OUTPUT

```
32500.0 <class 'float'>
37.5 <class 'float'>
0.125 <class 'float'>
True <class 'bool'>
TM0042
Bonus: 4,062.50 cr
```

Exercise 5 Yes or no

5 min · `ex5_true_false.py`

> **Raven from Lady Siv (Rewards):** *"Just yes or no, please. Dalvric's version once answered 'mayhap'."*

GIVEN

```
weekly_hours = 30
months_of_service = 14
rating = 4
salary = 33500
band_min = 25000
band_max = 32000
```

(a) Print four answers. **Full time**: 35 hours or more. **Bonus eligible**: at least 12 months *and* rating 3 or more. **Within band**: salary between min and max, limits included. **Needs review**: not within band.

(b) Before running it again: if `weekly_hours = 37.5` and `salary = 30000`, what will it print? Write your guess as a comment at the bottom of the file, then try it.

EXPECTED OUTPUT

```
Full time: False
Bonus eligible: True
Within band: False
Needs review: True
```

Exercise 6 LEFT, RIGHT, MID

10 min · `ex6_slicing.py`

> **Raven from Orrin:** *"In Excel I'd use LEFT/MID/RIGHT. Dalvric used four hundred lines. Please use fewer."*

GIVEN

```
realm = "Thornmere"
ref = "GUARD-1203"
```

(a) Print the Python versions of `LEFT(realm, 5)`, `RIGHT(realm, 4)` and `MID(realm, 3, 4)`.

(b) Split `ref` at the dash: print the guild, then the number as a real number.

(c) Print the number plus one, to prove it really is a number now.

EXPECTED OUTPUT

```
Thorn
mere
ornm
GUARD
1203
1204
```

Hint: Excel counts from 1, Python from 0. `ref.find("-")` tells you where the dash is. Your code should still work if the ref were `"FORGE-0007"`. Try it!

Exercise 7 The pay gap in one sentence STRETCH

15 min · `ex7_pay_gap.py`

> **Decree from Queen Aldara:** *"The Privy Council wants our pay gap 'in one sentence a queen can say out loud'. Council sits Thursday. No pressure. (Some pressure.) P.S. Biscuit's flight log is in decimal hours and nobody understands it."*

GIVEN

```
male_hourly = 20.00
female_hourly = 17.40
flight_hours = 1.75
```

The Accord's definition: **gap % = (male − female) / male × 100**, rounded to 1 decimal. Pennies per crown = female / male × 100, rounded to a whole number.

EXPECTED OUTPUT

```
Women earn 87 pennies for every crown that men earn (gap: 13.0%).
Biscuit flew for 1h 45m.
```

Hint: `int(1.75)` is `1`. The minutes come from what's left over. `{minutes:02d}` pads to two digits.

Bug hunt Pages from the Grimoire, page 1

15 min · `bughunt/payroll_bug.py`

Dalvric translated a corner of the Grimoire into Python and sailed west. Since then the raise calculation crashes, and for three months nobody in the realm paid any crown tax (the Royal Auditors have questions). There are **three** bugs: one crashes, one gives a wrong number silently, and one you'll only spot by comparing letter by letter.

EXPECTED OUTPUT ONCE FIXED

```
Salary after 2 years: 31,827.00 cr
Osric Kettleby: gross 2345.50 cr, tax 469.10 cr, net 1876.40 cr
```

Commit each fix separately, with a message explaining what was wrong, like a detective's report.